CLEARANCE HULLS AND CONVERSE FAILURES FOR OKA COMPLEMENTS

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ABSTRACT. We show that for every nonempty compact regular closed set $K \subset \mathbb{C}^n$, $n \geq 2$,

$$H(K) = K \implies \mathbb{C}^n \setminus K \text{ is Oka} \implies H^-(K) = K,$$

where $H(K)$ records the points through which no nonconstant entire curve stays a positive distance from K, and $H^-(K)$ is its inner counterpart formed from the clearance hulls of compact subsets of $\operatorname{int} K$. We give counterexamples to both converse implications. As an application, we give closures of bounded domains with smooth real-algebraic boundaries whose complements are Oka by the clearance criterion although the available polynomial-convexity, tangential Levi-form, and periodic-transport results do not apply.

1. INTRODUCTION

The classical Oka principle has its roots in Oka's 1939 solution of the second Cousin problem on domains of holomorphy [18]. Grauert's work in the 1950s established the Oka–Grauert principle for holomorphic fiber bundles over Stein spaces [14]. The decisive nonlinear step came with Gromov's 1989 theory of elliptic manifolds: dominating sprays provide the geometric flexibility from which approximation, interpolation, and homotopy principles follow [15]. The subject thereby developed from a collection of extension theorems into a theory of flexible target manifolds.

The modern Oka property packages these forms of flexibility into an equivalent family of approximation and interpolation principles. We use its standard convex-approximation formulation: a complex manifold Y is Oka if, for every integer $m \geq 1$, every compact convex set $Q \subset \mathbb{C}^m$, every holomorphic map $f : U \to Y$ from an open neighborhood $U \subset \mathbb{C}^m$ of Q, and every $\epsilon > 0$, there is an entire map $F : \mathbb{C}^m \to Y$ such that

$$\sup_{z \in Q} d_Y(F(z), f(z)) < \epsilon,$$

where d_Y is any distance inducing the topology of Y. Forstnerič [8, Theorem 0.1] proved that this convex approximation property is equivalent to the classical Oka property; the full family of equivalent properties and their parametric versions is presented systematically in [9, Chapter 5]. Kusakabe's convex-ellipticity theorem later gave a geometric characterization of Oka manifolds and Oka maps [16, Theorem 1.3]. The homotopy-theoretic formulation of Oka theory, surveyed by Forstnerič and Lárusson [10], provides a broader conceptual framework but will not be needed here. One consequence that is central here is that every point of a positive-dimensional Oka manifold lies on a nonconstant entire curve [7, Lemma 2.1].

We use the weaker Oka-1 property only for the second counterexample. A complex manifold Y is Oka-1 if each of its connected components has the following property: for every open Riemann surface R, every Runge compact set $A \subset R$ (equivalently, $R \setminus A$ has no relatively compact connected component), every discrete sequence (a_i) in R without repetitions, every continuous map $f : R \to Y$ that is holomorphic near $A \cup \{a_i : i \geq 1\}$,

Key words and phrases. Oka manifold, clearance hull, entire curve, compact complement, polynomial convexity, Levi form.

every $\epsilon > 0$, and every sequence of positive integers (k_i), there is a holomorphic map $F : R \to Y$, homotopic to f, such that

$$\sup_{x \in A} d_Y(F(x), f(x)) < \epsilon$$

for a distance d_Y inducing the topology of Y, and F agrees with f to order k_i at a_i for every i [1, Definition 1.1].

One basic problem is to determine which closed subsets can be removed from a flexible ambient manifold without destroying the Oka property. Direct spray constructions on complements are difficult, even for compact convex sets in Euclidean space. Forstnerič and Ritter proved approximation and interpolation for maps from Stein sources into complements of compact convex sets, subject to dimension bounds, and obtained a corresponding result for polynomially convex sets under a stronger bound [11]. These results anticipated, but did not by themselves establish, the full Oka property of every such complement. The long-standing polynomially convex complement problem was settled by Kusakabe. Recall that a Stein manifold M has the density property if the Lie algebra generated by its complete holomorphic vector fields is dense, in the compact-open topology, in the Lie algebra of all holomorphic vector fields. A compact set $K \subset M$ is $\mathscr{O}(M)$-convex if, for every $p \in M \setminus K$, some $h \in \mathscr{O}(M)$ satisfies $|h(p)| > \max_K |h|$. If M is a Stein manifold with the density property and $K \Subset M$ is $\mathscr{O}(M)$-convex, then $M \setminus K$ is Oka [17, Theorem 1.2]. In particular, every compact polynomially convex set in $\mathbb{C}^n$, $n \geq 2$, has Oka complement [17, Corollary 1.3]. Forstnerič and Wold obtained another route through holomorphic families of Fatou–Bieberbach domains, that is, proper subdomains of $\mathbb{C}^n$ biholomorphic to $\mathbb{C}^n$, [12, Theorem 1.1]. Their later work treats several unbounded convex and projectively convex sets [13, Theorem 1.1 and Corollary 1.5]. Du, Guo, Wang, and Xie subsequently used periodic divergence-free transport and attracting basins to prove that the complement of a $2\pi\mathbb{Z}^n$-periodic closed set whose quotient in $(\mathbb{C}^*)^n$ is compact and holomorphically convex is Oka. Their applications include convex tubes and certain Cartesian products of planar compact sets [6, Theorems 1.1 and 1.2].

Convexity is not the only geometry that can force an Oka complement. For a real C^2 function ρ on a domain in $\mathbb{C}^n$, our convention for its Levi form is

$$\mathcal{L}_\rho(p; v) = \sum_{j,k=1}^{n} \rho_{z_j \bar{z}_k}(p) v_j \bar{v}_k.$$

If $K = \{\rho \leq 0\}$ and $d\rho \neq 0$ on ∂K, the tangential Levi form is the restriction of $\mathcal{L}_\rho$ to $\ker \partial\rho_p = T^{1,0}_p \partial K$. For smooth compact domains, Du [7, Theorem 1.1] gives a criterion in terms of the Levi form restricted to the complex tangent bundle: the complement is Oka if this Hermitian form has at least two nonnegative eigenvalues at every boundary point of a compact domain in a Stein manifold with the density property. Polynomial convexity is a global condition and can fail in the presence of a single analytic disc, whereas the cited Levi-form condition is local and requires a smooth boundary. The condition introduced here is global, applies to arbitrary compact sets, and is expressed solely in terms of the Euclidean distance between the compact set and images of entire curves.

For a nonempty compact set $K \subset \mathbb{C}^n$, the *clearance function* is

$$e_K(p) = \sup_f \operatorname{dist}(f(\mathbb{C}), K), \qquad p \in \mathbb{C}^n, \tag{1.1}$$

where $f : \mathbb{C} \to \mathbb{C}^n$ ranges over the nonconstant entire maps satisfying $f(0) = p$. Its zero set is the *clearance hull* $H(K)$. If K is regular closed, meaning $K = \overline{\operatorname{int} K}$, its *inner clearance hull* is denoted by $H^-(K)$; explicitly,

$$H(K) = \{p : e_K(p) = 0\}, \qquad H^-(K) = \overline{\bigcup_{\varnothing \neq L \Subset \operatorname{int} K} H(L)}. \tag{1.2}$$

Thus $p \notin H(K)$ precisely when there is a nonconstant entire map $f : \mathbb{C} \to \mathbb{C}^n$ with $f(0) = p$ and $\operatorname{dist}(f(\mathbb{C}), K) > 0$. The set $H^-(K)$ is obtained by taking this zero set for every nonempty compact $L \Subset \operatorname{int} K$, forming their union, and then taking its closure in $\mathbb{C}^n$.

Theorem 1.1. *Let $K \subset \mathbb{C}^n$, $n \geq 2$, be a nonempty compact regular closed set. Then*

$$H(K) = K \implies \mathbb{C}^n \setminus K \text{ is Oka} \implies H^-(K) = K. \tag{1.3}$$

Both reverse implications fail. More precisely:

(1) *for every $n \geq 2$ there is a compact regular closed set with connected interior and C^∞ boundary whose complement is Oka but whose clearance hull strictly contains it;*
(2) *for every $n \geq 2$ there is a connected compact regular closed set K with connected complement such that $H^-(K) = K$ while $\mathbb{C}^n \setminus K$ is neither Oka nor Oka-1.*

We now describe the ideas of the proof. For the first implication, suppose that $e_K(p) > \delta$. An entire curve through p whose image has distance greater than δ from K remains at distance greater than δ after sufficiently small translations and after applying linear maps sufficiently close to the identity. Choosing n such linear maps gives holomorphic families whose derivatives in the curve parameter form a basis of $T_p\mathbb{C}^n$. The local approximation criterion of Du, Guo, Wang, and Xie [5, Theorem 2.2, Lemma 2.3, and Proposition 5.2] then shows that

$$C_\delta(K) = \{p \in \mathbb{C}^n : e_K(p) > \delta\}$$

is Oka for every $\delta \geq 0$. In particular, $C_0(K) = \mathbb{C}^n \setminus H(K)$ is Oka. No global spray on this complement is constructed.

For the second implication, [7, Lemma 2.1] gives, for every $p \in X = \mathbb{C}^n \setminus K$, a nonconstant entire curve $f : \mathbb{C} \to X$ with $f(0) = p$. If $L \Subset \operatorname{int} K$, then $\operatorname{dist}(f(\mathbb{C}), L) \geq \operatorname{dist}(X, L) > 0$, and hence $p \notin H(L)$. Choose $r > 0$ such that the closed r-neighborhood L_r is contained in $\operatorname{int} K$, where $L_r = \{z \in \mathbb{C}^n : \operatorname{dist}(z, L) \leq r\}$. Proposition 2.6 gives

$$H(L) \subset \{e_L < r\} \subset H(L_r) \subset K,$$

so the open set $\{e_L < r\}$ is a neighborhood of $H(L)$ contained in K, and hence $H(L) \subset \operatorname{int} K$. Formula (1.2) then gives $H^-(K) = K$. The converse may fail because this argument treats each compact set $L \Subset \operatorname{int} K$ separately and gives no uniform positive lower bound as L approaches ∂K.

The first counterexample is a smooth worm-type compact set. Its complement is Oka by propagation of the local property $\mathcal{L}_n$ across coordinate-line discs, while the continuity principle for analytic hypersurfaces implies that a specified point of the complement belongs to the clearance hull. For the second counterexample, every nonempty inner parallel set $K_{-\delta} = \{z \in K : \operatorname{dist}(z, \mathbb{C}^n \setminus K) \geq \delta\}$ is polynomially convex and hence has Oka complement. Nevertheless, the limiting complement admits a bounded plurisubharmonic function u and a nonempty open set W such that every entire curve $f : \mathbb{C} \to \mathbb{C}^n \setminus K$ meeting W is constant. Interpolation at two points then shows directly that the complement is not Oka-1.

Finally, the real-algebraic compact set in Theorem 3.3 shows how the clearance criterion can be used when the established sufficient conditions are unavailable. If $H(K) \setminus K$ were nonempty, the defining polynomial would have a positive maximum on $H(K)$; [7, Lemma 2.5] and Lemma 2.4 exclude such a maximum. This proves $H(K) = K$ without a spray construction. The compact set is not polynomially convex, and along a specified boundary circle its tangential Levi form has only one nonnegative eigenvalue, so the hypothesis of [7, Theorem 1.1] is not satisfied. The example therefore distinguishes the clearance condition from the polynomial-convexity and Levi-form criteria. It is also outside the periodic and product classes in [6, Theorems 1.1 and 1.2], as verified after Theorem 3.3.

Section 2 develops the clearance hull and proves its criterion for Oka complements, together with the necessary inner clearance condition. Section 3 constructs a smooth worm-type compact domain whose Oka complement does not imply $H(K) = K$, and then applies the criterion to a real-algebraic compact domain outside the previously available sufficient results. Section 4 constructs a regular closed set with polynomially convex inner parallel sets but non-Oka complement, and records the remaining smooth-boundary question.

2. Clearance Hulls and Oka Complements

Throughout the paper, all norms and distances in Euclidean spaces are the standard Euclidean ones. In this section, $n \geq 2$. Write $B(a,r) = \{z \in \mathbb{C}^n : \|z-a\| < r\}$ and $B_r = B(0,r)$. For discs in $\mathbb{C}$, write $\Delta(a,r) = \{\zeta \in \mathbb{C} : |\zeta - a| < r\}$, $\Delta_r = \Delta(0,r)$, and $\Delta = \Delta_1$. For nonempty compact sets $K, L \subset \mathbb{C}^n$, $d_H(K,L)$ denotes their Hausdorff distance. The polynomial hull of a compact set K is

$$\widehat{K} = \{z \in \mathbb{C}^n : |P(z)| \leq \max_K |P| \text{ for every } P \in \mathbb{C}[z_1, \ldots, z_n]\}.$$

Let d_Y be any distance inducing the topology of an n-dimensional complex manifold Y. A *special planar disc pair* $D \subset D'$ consists of compact topological discs in $\mathbb{C}$ with piecewise smooth boundaries, where D' is obtained by adjoining a disc to D along a proper boundary arc. Following [5, Definition 2.1], we say that Y has property $\mathcal{L}_n$ at $y \in Y$ if there is an open neighborhood V of y such that, for every special planar disc pair $D \subset D'$, every $m \geq 0$, all closed polydiscs $P_0 \Subset \operatorname{int} P \subset \mathbb{C}^m$, and every V-valued map f holomorphic on an open neighborhood of $D \times P$, there are maps

$$F_\nu : U_\nu \longrightarrow Y, \qquad D' \times P_0 \subset U_\nu \subset \mathbb{C}^{m+1} \text{ open}, \quad \text{such that} \quad \sup_{D \times P_0} d_Y(F_\nu, f) \longrightarrow 0.$$

For $m = 0$, both polydiscs are the one-point space $\mathbb{C}^0$. The definition is independent of the choice of the compatible distance d_Y. Write

$$\mathcal{G}(Y) = \{y \in Y : Y \text{ has property } \mathcal{L}_n \text{ at } y\}.$$

The following consequences of [5, Theorem 2.2, Lemma 2.3, Propositions 5.2–5.3] will be used:

(i) The set $\mathcal{G}(Y)$ is open, and $\mathcal{G}(Y) = Y$ implies that Y is Oka.
(ii) If $F : \mathbb{C}^n \to Y$ is holomorphic, $F(0) = y$, and $dF_0 : \mathbb{C}^n \to T_yY$ has rank n, then $y \in \mathcal{G}(Y)$.
(iii) Suppose that $V \subset Y$ is a neighborhood of p and $h_j : V \times \mathbb{C} \to Y$, $1 \leq j \leq n$, are holomorphic maps satisfying $h_j(y,0) = y$ for every $y \in V$. If the vectors $\partial_t h_j(p,0)$ form a basis of T_pY, then $p \in \mathcal{G}(Y)$.
(iv) Let $\vartheta : U \to \Omega \subset \mathbb{C}^n$ be a holomorphic coordinate chart. Fix $p \in U$ and $v \in \mathbb{C}^n \setminus \{0\}$. If, for some $r > 0$,

$$\{\vartheta(p) + \zeta v : |\zeta| \leq r\} \subset \Omega \quad \text{and} \quad \vartheta^{-1}\{\vartheta(p) + \zeta v : |\zeta| = r\} \subset \mathcal{G}(Y),$$

then $p \in \mathcal{G}(Y)$.

In (iii), Proposition 5.2 is applied to the restrictions of the maps h_j to discs whose radii tend to infinity. If $U \subset Y$ is open and is Oka as a complex manifold, then $U \subset \mathcal{G}(Y)$ by [7, Lemma 2.2].

Proposition 2.1. *For nonempty compact sets $K, L \subset \mathbb{C}^n$, the clearance functions satisfy*

$$0 \leq e_K(p) \leq \operatorname{dist}(p, K), \tag{2.1}$$

$$|e_K(p) - e_K(q)| \leq \|p - q\|, \tag{2.2}$$

$$\sup_{p \in \mathbb{C}^n} |e_K(p) - e_L(p)| \leq d_H(K, L). \tag{2.3}$$

The hull $H(K)$ is compact, and $K \subset L$ implies $H(K) \subset H(L)$. For every biholomorphic automorphism Φ of $\mathbb{C}^n$,

$$H(\Phi(K)) = \Phi(H(K)).$$

Moreover,

$$e_{H(K)} = e_K, \qquad H(H(K)) = H(K), \tag{2.4}$$

and, for $r \geq 0$, $K_r = \{z : \operatorname{dist}(z, K) \leq r\}$ and $x_+ = \max\{x, 0\}$,

$$e_{K_r} = (e_K - r)_+, \qquad H(K_r) = \{e_K \leq r\}. \tag{2.5}$$

Proof. If $f(0) = p$, then $p \in f(\mathbb{C})$, which proves (2.1). For every nonconstant entire map $f : \mathbb{C} \to \mathbb{C}^n$ with $f(0) = p$, the translated map $f + q - p$ takes 0 to q, and

$$\big|\operatorname{dist}(f(\mathbb{C}) + q - p, K) - \operatorname{dist}(f(\mathbb{C}), K)\big| \leq \|p - q\|.$$

Taking the supremum over f and then interchanging p and q gives (2.2). For every nonempty set A,

$$|\operatorname{dist}(A, K) - \operatorname{dist}(A, L)| \leq d_H(K, L).$$

Taking the supremum over all nonconstant entire maps through p gives (2.3). If $K \subset L$, then $e_L \leq e_K$, so $H(K) \subset H(L)$.

Choose R with $K \subset B_R$. If $\|p\| > R$, there is a nonzero vector $v \perp p$, since $n \geq 2$, and

$$\|p + \zeta v\|^2 = \|p\|^2 + |\zeta|^2 \|v\|^2.$$

This affine line has distance at least $\|p\| - R$ from K. Thus $H(K) \subset \overline{B_R}$; it is closed by (2.2). For every subset $A \subset \mathbb{C}^n$ and compact set K, one has $\operatorname{dist}(A, K) > 0$ if and only if $\overline{A} \cap K = \varnothing$. A homeomorphism preserves disjointness and satisfies $\Phi(\overline{A}) = \overline{\Phi(A)}$. Applying these facts to $A = f(\mathbb{C})$ and to Φ and Φ^{-1} proves the automorphism assertion.

For a nonconstant entire map $f : \mathbb{C} \to \mathbb{C}^n$, write $d = \operatorname{dist}(f(\mathbb{C}), K)$. Recentring its source gives $e_K(f(t)) \geq d$ at every t. Since e_K vanishes on $H(K)$, (2.2) yields

$$\operatorname{dist}(f(t), H(K)) \geq e_K(f(t)) \geq d.$$

Infima over t and suprema over f give $e_{H(K)} \geq e_K$. The opposite inequality follows from $K \subset H(K)$. This proves (2.4).

Finally,

$$\operatorname{dist}(z, K_r) = (\operatorname{dist}(z, K) - r)_+.$$

The positive-part function is increasing and continuous, so it commutes with the infimum over an image and the supremum in (1.1). This proves (2.5). □

Theorem 2.2. *Let $K \subset \mathbb{C}^n$ be a nonempty compact set. For every $\delta \geq 0$, the open set*

$$C_\delta(K) = \{p : e_K(p) > \delta\}$$

is connected and Oka. If $\delta > 0$, then $\operatorname{dist}(C_\delta(K), K) \geq \delta$. In particular, $\mathbb{C}^n \setminus H(K)$ is Oka.

Proof. Fix $p \in C_\delta(K)$ and a nonconstant entire curve f through p such that $d = \operatorname{dist}(f(\mathbb{C}), K) > \delta$. Since f is nonconstant, the zeros of f' are isolated; hence $f'(s_\nu) \neq 0$ for a sequence $s_\nu \to 0$. The maps

$$t \longmapsto f(t + s_\nu) + p - f(s_\nu)$$

take 0 to p, and their image distances from K tend to at least d as $\nu \to \infty$. Replacing f by one of these maps gives $f'(0) \neq 0$ while preserving $\operatorname{dist}(f(\mathbb{C}), K) > \delta$.

For $A \in \mathrm{GL}_n(\mathbb{C})$, let $\Phi_A(z) = p + A(z - p)$. Compactness of K gives

$$d_H(\Phi_A^{-1}(K), K) \longrightarrow 0 \quad (A \to I).$$

If $\sigma_{\min}(A)$ is the least singular value of A, then

$$\operatorname{dist}(\Phi_A(f(\mathbb{C})), K) \geq \sigma_{\min}(A)\big(d - d_H(\Phi_A^{-1}(K), K)\big) > \delta$$

for A sufficiently close to I. Choose a basis $w_1, \dots, w_n$ of $\mathbb{C}^n$ with every w_j arbitrarily close to $f'(0)$. There are matrices $A_j \in \mathrm{GL}_n(\mathbb{C})$, arbitrarily close to I, such that $A_j f'(0) = w_j$. After choosing them close enough to I, the preceding inequality holds for every A_j.

One neighborhood $V \ni p$ makes all the holomorphic families

$$h_j(y,t) = y - p + \Phi_{A_j}(f(t)), \qquad (y,t) \in V \times \mathbb{C},$$

have image distance greater than δ from K. If $q = h_j(y, t_0)$, then the nonconstant entire map $t \mapsto h_j(y, t+t_0)$ takes 0 to q and has the same image as $h_j(y, \cdot)$; hence $q \in C_\delta(K)$. Thus each h_j maps $V \times \mathbb{C}$ into $C_\delta(K)$, satisfies $h_j(y,0) = y$, and has $\partial_t h_j(p,0) = A_j f'(0) = w_j$. Item (iii) above gives $\mathcal{G}(C_\delta(K)) = C_\delta(K)$, and item (i) proves that $C_\delta(K)$ is Oka.

The affine-line argument in Proposition 2.1 shows that $C_\delta(K)$ contains $\{\|z\| > R + \delta\}$ whenever $K \subset B_R$. For every $p \in C_\delta(K)$, choose a nonconstant entire map $f : \mathbb{C} \to \mathbb{C}^n$ with $f(0) = p$ and $\mathrm{dist}(f(\mathbb{C}), K) > \delta$. Its image is unbounded, since otherwise all of its coordinate functions would be bounded and hence constant. For every $t_0 \in \mathbb{C}$, the map $t \mapsto f(t + t_0)$ shows that $f(t_0) \in C_\delta(K)$. Thus $f(\mathbb{C}) \subset C_\delta(K)$. The image is connected and meets $\{\|z\| > R + \delta\}$, so every connected component of $C_\delta(K)$ meets this same connected set. Hence $C_\delta(K)$ is connected. Finally, (2.1) gives $\mathrm{dist}(p, K) > \delta$ for every $p \in C_\delta(K)$ and therefore $\mathrm{dist}(C_\delta(K), K) \geq \delta$. □

The first implication of (1.3) follows immediately. Two descriptions of its strength will be useful.

Corollary 2.3. *For every nonempty compact set $K \subset \mathbb{C}^n$, the set $\mathbb{C}^n \setminus H(K)$ is the union of all Oka open subsets of $\mathbb{C}^n \setminus K$ whose distance from K is positive. Also*

$$K \subset H(K) \subset \widehat{K}.$$

Proof. The equality $\mathbb{C}^n \setminus H(K) = \bigcup_{j \geq 1} C_{1/j}(K)$ and Theorem 2.2 give one inclusion in the union assertion. Conversely, suppose that $U \subset \mathbb{C}^n \setminus K$ is open and Oka and that $\mathrm{dist}(U, K) > 0$. By [7, Lemma 2.1], every $p \in U$ lies on a nonconstant entire curve $f : \mathbb{C} \to U$. Then $\mathrm{dist}(f(\mathbb{C}), K) \geq \mathrm{dist}(U, K) > 0$, so $p \notin H(K)$.

If $p \notin \widehat{K}$, polynomial separation gives a compact polynomial polyhedron L with $K \subset \mathrm{int}\, L$ and $p \notin L$. Indeed, choose a polynomial P and a number c such that

$$\max_K |P| < c < |P(p)|,$$

and choose $R > \max_{z \in K} \|z\|$. Then

$$L = \{z \in \mathbb{C}^n : |P(z)| \leq c,\ |z_j| \leq R\ (1 \leq j \leq n)\}$$

has the required properties. It is polynomially convex: if $q \notin L$, then either $|P(q)| > c \geq \max_L |P|$ or $|q_j| > R \geq \max_L |z_j|$ for some j, so a polynomial separates q from L. The complement of L is Oka by [17, Corollary 1.3]. A nonconstant entire map through p with image in that complement has distance at least $\mathrm{dist}(K, \mathbb{C}^n \setminus \mathrm{int}\, L) > 0$ from K. Thus $p \notin H(K)$. □

Lemma 2.4. *Suppose that $K \subset \mathbb{C}^n$ is nonempty and compact. Let $\vartheta : U \to \Omega \subset \mathbb{C}^n$ be a holomorphic coordinate chart. Fix $p \in U$, $v \in \mathbb{C}^n \setminus \{0\}$, and $r > 0$, and suppose that*

$$D = \vartheta^{-1}\{\vartheta(p) + \zeta v : |\zeta| \leq r\} \Subset \mathbb{C}^n \setminus K.$$

If the boundary of D is contained in $\mathbb{C}^n \setminus H(K)$, then $D \subset \mathbb{C}^n \setminus H(K)$.

Proof. Since the boundary of D is compact and $\mathbb{C}^n \setminus H(K) = \bigcup_{j \geq 1} C_{1/j}(K)$, there is a $\delta > 0$ such that $\partial D \subset C_\delta(K)$. After an affine change in the coordinate range, the disc is $\overline{\Delta_r} \times \{0\} \subset \mathbb{C} \times \mathbb{C}^{n-1}$. There are $\epsilon, \eta > 0$ such that

$$\vartheta^{-1}(\overline{\Delta_{r+\epsilon}} \times \overline{\Delta_\eta}^{n-1}) \Subset \mathbb{C}^n \setminus K$$

and

$$\vartheta^{-1}\big(\{r-\epsilon\le|\zeta|\le r+\epsilon\}\times\overline{\Delta_\eta}^{n-1}\big)\subset C_\delta(K).$$

Let

$$T=\vartheta^{-1}(\Delta_{r+\epsilon}\times\Delta_{\eta/2}^{n-1}),\qquad Z=C_\delta(K)\cup T.$$

The set Z is open and $\operatorname{dist}(Z,K)>0$. Since $C_\delta(K)$ is Oka, it is contained in $\mathcal{G}(Z)$. Write a point of $\vartheta(T)$ as (z_1,z'), set $R=r+\epsilon$, and choose an automorphism ψ of Δ with $R\psi(0)=z_1$. The functions $t\mapsto|R\psi(\rho e^{it})|$ converge uniformly to R as $\rho\uparrow 1$. Hence, for $\rho<1$ sufficiently close to 1,

$$\zeta\longmapsto\vartheta^{-1}(R\psi(\rho\zeta),z')$$

is a closed coordinate-line disc after the holomorphic change of first coordinate induced by ψ; it is contained in T, and its boundary is contained in $C_\delta(K)\subset\mathcal{G}(Z)$. Item (iv) above therefore gives $T\subset\mathcal{G}(Z)$. Hence $\mathcal{G}(Z)=Z$, item (i) shows that Z is Oka, and Corollary 2.3 yields $Z\subset\mathbb{C}^n\setminus H(K)$. Since $D\subset T$, the conclusion follows. □

Theorem 2.5. *For an open set $D\subset\mathbb{C}^n$, the set*

$$U_D=\bigcup_{\varnothing\neq L\Subset D}H(L)$$

is open and satisfies

$$H(B)\subset U_D\qquad\text{for every nonempty compact set } B\Subset U_D. \tag{2.6}$$

It is the smallest open set V containing D such that $H(B)\subset V$ for every nonempty compact $B\Subset V$. For a compact regular closed set K,

$$H^-(K)=K\quad\Longleftrightarrow\quad H(L)\subset\operatorname{int}K\ \text{ for every nonempty compact } L\Subset\operatorname{int}K. \tag{2.7}$$

Proof. Given $L\Subset D$, choose $r>0$ with $L_r\Subset D$. Equation (2.5) gives

$$H(L)\subset\{e_L<r\}\subset H(L_r)\subset U_D.$$

The middle open set is a neighborhood of $H(L)$, so U_D is open. For a compact set $B\Subset U_D$, choose finitely many compact sets $L_i\Subset D$ and numbers $r_i>0$ such that $(L_i)_{r_i}\Subset D$ and the corresponding open sets $\{e_{L_i}<r_i\}$ cover B. The compact set $L=\bigcup_i(L_i)_{r_i}$ lies in D, and monotonicity gives $B\subset H(L)$. Hence

$$H(B)\subset H(H(L))=H(L)\subset U_D.$$

This proves (2.6). Since $L\subset H(L)$, the set U_D contains D. If an open set $V\supset D$ satisfies $H(B)\subset V$ for every nonempty compact $B\Subset V$, then it contains $H(L)$ for every nonempty compact $L\Subset D$ and hence $U_D\subset V$. This proves the minimality assertion.

If $H^-(K)=K$, then $H(L_r)\subset K$ for every $L_r\Subset\operatorname{int}K$. The open neighborhood $\{e_L<r\}$ of $H(L)$ is contained in $H(L_r)\subset K$, so $H(L)\subset\operatorname{int}K$. Conversely, suppose that $H(L)\subset\operatorname{int}K$ for every nonempty compact $L\Subset\operatorname{int}K$. Then $U_{\operatorname{int}K}\subset\operatorname{int}K$. The reverse inclusion follows from $L\subset H(L)$, applied to one-point sets $L\subset\operatorname{int}K$. Thus $U_{\operatorname{int}K}=\operatorname{int}K$, and taking closures gives $H^-(K)=\overline{\operatorname{int}K}=K$. □

Proposition 2.6. *Suppose that $K\subset\mathbb{C}^n$ is compact and regular closed, and write $X=\mathbb{C}^n\setminus K$. If every point of X lies on a nonconstant entire curve in X, then $H^-(K)=K$.*

Proof. Let $L\Subset\operatorname{int}K$ be nonempty, and choose $r>0$ such that $L_r\Subset\operatorname{int}K$. For $p\in X$, a nonconstant entire map $f:\mathbb{C}\to X$ with $f(0)=p$ satisfies

$$\operatorname{dist}(f(\mathbb{C}),L_r)\ge\operatorname{dist}(X,L_r)>0.$$

Hence $p\notin H(L_r)$ and therefore $H(L_r)\subset K$. By (2.5),

$$H(L)\subset\{e_L<r\}\subset H(L_r)\subset K.$$

The middle open set is a neighborhood of $H(L)$, so $H(L)\subset\operatorname{int}K$. Equation (2.7) now gives $H^-(K)=K$.

If X is Oka, then [7, Lemma 2.1] supplies the required nonconstant entire curve through every $p \in X$. This proves the second implication in (1.3). □

3. Worm-Type and Real-Algebraic Compact Sets with Oka Complements

3.1. **The Worm-Type Compact Set.** The counterexample is a variant of the worm construction of Diederich and Fornæss [4]. It uses the following explicit smooth convex cutoff. Let

$$\chi(s) = \begin{cases} 0, & s \le 0, \\ \exp(s - s^{-1}), & s > 0, \end{cases} \qquad \eta_\mu(\theta) = \chi(\theta - \mu) + \chi(-\theta - \mu), \quad \mu > 0.$$

The function χ is C^∞, with all right derivatives zero at 0, and, for $s > 0$,

$$\chi'(s) = \chi(s)(1 + s^{-2}), \qquad \chi''(s) = \frac{\chi(s)}{s^4}\big(s^4 + 2s^2 - 2s + 1\big) > 0.$$

The last numerator is positive because $2s^2 - 2s + 1 = 2(s - \frac{1}{2})^2 + \frac{1}{2}$. Thus η_μ is convex, vanishes on $[-\mu, \mu]$, and equals 1 exactly at $\theta = \pm(\mu + 1)$.

For $(a, b) \in \mathbb{C} \times (\mathbb{C}^{n-1} \setminus \{0\})$, let the logarithmic radial coordinate be $\theta = \log\|b\|^2$ and define

$$\rho_\mu(a, b) = |a - e^{i\theta}|^2 - 1 + \eta_\mu(\theta),$$

Set

$$K_\mu = \{(a, b) \in \mathbb{C} \times (\mathbb{C}^{n-1} \setminus \{0\}) : \rho_\mu(a, b) \le 0\}.$$

Theorem 3.1. *The following assertions hold.*

(i) *For every $n \ge 2$ and $\mu > 0$, the set K_μ is the closure of a bounded connected domain with C^∞ boundary. At every point of ∂K_μ, the Levi form of ρ_μ restricted to the complex tangent space has at least one nonnegative eigenvalue, and $X_\mu = \mathbb{C}^n \setminus K_\mu$ is Oka.*

(ii) *For $\mu = \pi$ and $p = (-\frac{1}{2}, e_1)$, where $e_1 = (1, 0, \dots, 0) \in \mathbb{C}^{n-1}$,*

$$p \in H(K_\pi) \setminus K_\pi, \qquad H^-(K_\pi) = K_\pi.$$

Proof of Theorem 3.1(i). *Step 1: Geometry of the compact set.* Membership in K_μ implies

$$e^{-(\mu+1)/2} \le \|b\| \le e^{(\mu+1)/2}, \qquad |a| \le 2.$$

Thus K_μ is compact and stays away from $b = 0$. If $\partial\rho_\mu/\partial a = 0$, then $a = e^{i\theta}$. At a point of the zero level set this implies $\eta_\mu(\theta) = 1$, and hence $\eta'_\mu(\theta) \ne 0$. At such a point, $d\rho_\mu = \eta'_\mu(\theta)d\theta \ne 0$. Therefore 0 is a regular value of ρ_μ. Therefore ∂K_μ is smooth and $K_\mu = \overline{\operatorname{int} K_\mu}$. The set $\{b \in \mathbb{C}^{n-1} : |\log\|b\|^2| < \mu + 1\}$ is connected. For each such b, the a-fiber of $\operatorname{int} K_\mu$ is the disc

$$\{a \in \mathbb{C} : |a - e^{i\theta}|^2 < 1 - \eta_\mu(\theta)\}.$$

Radial contraction of each disc to its center $e^{i\theta}$, followed by a path in the displayed set of b-values, proves that $\operatorname{int} K_\mu$ is connected.

At a point with $b = b_0 \ne 0$, restrict to the holomorphic two-dimensional slice $b = e^t b_0$, $t = x + i\sigma$, and write $\theta_0 = \log\|b_0\|^2$. For a tangent vector $\nu = (\nu_a, \nu_t)$ to this slice, multiplication of ρ_μ by the positive function $e^{2\sigma}$ gives

$$e^{2\sigma}\rho_\mu(a, e^t b_0) = |ae^{-it}|^2 - 2\,\mathrm{Re}\big(ae^{-i\theta_0}e^{-2it}\big) + e^{2\sigma}\eta_\mu(\theta_0 + 2x),$$
$$\mathcal{L}_{e^{2\sigma}\rho_\mu}(\nu) = |d(ae^{-it})(\nu)|^2 + e^{2\sigma}(\eta_\mu + \eta''_\mu)(\theta_0 + 2x)|\nu_t|^2 \ge 0. \tag{3.1}$$

The tangent space to the complex two-dimensional slice intersects the complex tangent hyperplane of ∂K_μ in a complex subspace of dimension at least one. On the complex tangent space, multiplication of a defining function by a positive function multiplies its Levi form by that function. Since (3.1) is nonnegative on the whole slice, its Rayleigh quotient

is nonnegative on this intersection. Therefore the largest eigenvalue of the tangential Levi form is nonnegative.

Step 2: Points in the $\mathcal{L}_n$-locus.

We claim that every point of $X_\mu = \mathbb{C}^n \setminus K_\mu$ satisfying

$$Q(a,b) := |a|^2 - 2\,\mathrm{Re}(ae^{-i\theta}) > 0, \qquad b \neq 0,$$

belongs to $\mathcal{G}(X_\mu)$. Fix $p = (a_p, b_p)$ in this set. Then $a_p \neq 0$ and

$$\theta_p = \log\|b_p\|^2, \qquad u_p = e^{i\theta_p}/a_p, \qquad \mathrm{Re}\, u_p < \tfrac{1}{2}.$$

For $c > 0$, consider

$$g_c(Y) = \tfrac{1}{2}e^{-2Y} + cY - \mathrm{Re}\, u_p.$$

Its minimum is

$$\min_{Y\in\mathbb{R}} g_c(Y) = \tfrac{c}{2}(1 - \log c) - \mathrm{Re}\, u_p. \tag{3.2}$$

There is a compact interval $I \Subset (0,\infty)$ containing 1 in its interior on which these minima are positive. The asymptotics at both ends of the real line then give a constant $\gamma > 0$ such that

$$g_c(Y) \geq \gamma(1 + |Y|), \qquad c \in I, \quad Y \in \mathbb{R}. \tag{3.3}$$

Indeed, as $Y \to +\infty$ the ratio by $1+|Y|$ tends uniformly to $c \geq \min I > 0$, and as $Y \to -\infty$ it tends uniformly to infinity; the remaining compact interval is covered by (3.2).

For each $c \in I$, the entire functions

$$u(\zeta) = u_p e^{\zeta}, \qquad t_c(\zeta) = \frac{u_p(e^{\zeta} - 1)}{ic}$$

satisfy $u(\zeta) = u_p + ict_c(\zeta) \neq 0$. For $\xi \in \mathbb{C}^{n-1}$ sufficiently close to 0 and $y = (a_y, b_y)$ in a sufficiently small neighborhood V of p, the map

$$h_{c,\xi}(y,\zeta) = \Big(a_y e^{2it_c(\zeta)-\zeta}, (b_y + \xi)e^{t_c(\zeta)} - \xi\Big) \tag{3.4}$$

is holomorphic on $V \times \mathbb{C}$ and satisfies $h_{c,\xi}(y,0) = y$. The neighborhood V and a neighborhood of 0 for ξ may be chosen independently of $c \in I$ so that

$$h_{c,\xi}(V \times \mathbb{C}) \subset X_\mu. \tag{3.5}$$

Choose $0 < m < M$ such that $m \leq \|b\| \leq M$ for every $(a,b) \in K_\mu$. If the second component $B = (b_y + \xi)e^t - \xi$ satisfies $m \leq \|B\| \leq M$, then, with $x = \mathrm{Re}\, t$,

$$m \leq e^x\|b_y + \xi\| + \|\xi\|, \qquad e^x\|b_y + \xi\| \leq M + \|\xi\|.$$

For y near p and $\|\xi\| < m/2$, these inequalities place x in a fixed interval $[-L, L]$. On that strip,

$$\Lambda(y,\xi,t) = \log\|(b_y + \xi) - \xi e^{-t}\|^2 - \theta_p$$

tends uniformly to zero as $y \to p$ and $\xi \to 0$, independently of $\mathrm{Im}\, t$. Write $A = a_y e^{2it-\zeta}$ and $B = (b_y + \xi)e^t - \xi$, where $t = t_c(\zeta)$. At (A,B), the condition $Q(A,B) > 0$ is equivalent to

$$\mathrm{Re}\left(\frac{a_p}{a_y}e^{i\Lambda(y,\xi,t)}(u_p + ict)\right) < \tfrac{1}{2}e^{-2\,\mathrm{Im}\, t}. \tag{3.6}$$

Indeed, the logarithmic radial coordinate of B is $\theta_p + 2x + \Lambda(y,\xi,t)$, and the identities $u_p + ict = u_p e^{\zeta}$ and $u_p = e^{i\theta_p}/a_p$ give

$$\frac{e^{i(\theta_p + 2x + \Lambda(y,\xi,t))}}{A} = e^{2\,\mathrm{Im}\, t}\frac{a_p}{a_y}e^{i\Lambda(y,\xi,t)}(u_p + ict).$$

Since $Q(A,B) > 0$ is equivalent to $\mathrm{Re}(e^{i\log\|B\|^2}/A) < 1/2$, this proves the equivalence with (3.6).

On $|x| \le L$, there is a constant $C_0 > 0$, independent of $c \in I$, such that $|u_p + ict| \le C_0(1 + |\operatorname{Im} t|)$. The difference between the left side of (3.6) and $\operatorname{Re} u_p - c \operatorname{Im} t$ is therefore bounded by $\epsilon C_0(1 + |\operatorname{Im} t|)$, where $\epsilon > 0$ can be made arbitrarily small by shrinking V and the neighborhood of 0 for ξ. Choose them so that $\epsilon C_0 < \gamma/2$. Equation (3.3) proves (3.6). If $\|B\| \notin [m, M]$, then $(A, B) \notin K_\mu$ by the choice of m and M. Thus (3.5) holds for every scalar parameter $\zeta \in \mathbb{C}$.

Since $t_c'(0) = -iu_p/c$, the derivative of $\zeta \mapsto h_{c,\xi}(p, \zeta)$ at $\zeta = 0$ is

$$D_{c,\xi} = (a_p(2u_p/c - 1),\ -iu_p(b_p + \xi)/c)\,.$$

Two distinct values $c_1, c_2 \in I$ with $\xi = 0$ give a basis of the two-plane generated by the a direction and $(0, b_p)$: multiplying $D_{c,0}$ by c and subtracting gives a nonzero pure a direction. Fix one of these two values of c. Replacing $\xi = 0$ by sufficiently small multiples of a basis of the Hermitian orthogonal complement $b_p^\perp \subset \mathbb{C}^{n-1}$ supplies the remaining $n - 2$ directions. Hence n maps in (3.4) have parameter derivatives at p that form a basis of T_pX_μ. Item (iii) in Section 2 proves the claim.

Step 3: Verification of the Oka property.

Choose $R > 0$ with $K_\mu \subset B_R$. Since $\overline{B_R}$ is polynomially convex, $\mathbb{C}^n \setminus \overline{B_R}$ is Oka by [17, Corollary 1.3] and hence is contained in $\mathcal{G}(X_\mu)$. If $p = (a, 0)$, the affine line $\zeta \mapsto (a + \zeta, 0)$ has positive distance from K_μ; hence a neighborhood of p lies in $C_0(K_\mu) \subset \mathcal{G}(X_\mu)$. Suppose that $S = X_\mu \setminus \mathcal{G}(X_\mu)$ is nonempty. It is relatively closed and bounded. Step 2 gives $Q \le 0$ on S; since $\rho_\mu = Q + \eta_\mu > 0$ on X_μ, one also has $\eta_\mu > 0$ on S. Fix $0 < \alpha < 1$. On S the positive function

$$F = \rho_\mu/\eta_\mu^\alpha$$

satisfies $F \le \eta_\mu^{1-\alpha}$. Its value tends to zero when a sequence in S approaches ∂K_μ, either by $\rho_\mu \to 0$ with $\eta_\mu > 0$ or by the displayed upper bound when $\eta_\mu \to 0$. The closure of S is disjoint from $\{b = 0\}$ because every point of this set has an open neighborhood in $\mathcal{G}(X_\mu)$. A maximizing sequence for $F|_S$ lies in a bounded set and therefore has a convergent subsequence. Its limit cannot belong to $X_\mu \setminus S = \mathcal{G}(X_\mu)$, since this set is open, and it cannot belong to ∂K_μ, since F tends to zero there. Consequently F attains a positive maximum C at some $p_* = (a_*, b_*) \in S$. Write $\theta_* = \log\|b_*\|^2$.

Near θ_*, let $\widetilde{\eta} = \eta_\mu - C\eta_\mu^\alpha$ and $\tau = -Q(p_*)/\eta_\mu(\theta_*) \in [0, 1)$. Then

$$\begin{aligned} C\eta_\mu^{\alpha-1}(\theta_*) &= 1 - \tau, \\ \widetilde{\eta}(\theta_*) &= \tau\eta_\mu(\theta_*) \ge 0, \\ \widetilde{\eta}''(\theta_*) &= [1 - \alpha(1 - \tau)]\eta_\mu''(\theta_*) + \alpha(1 - \alpha)(1 - \tau)\frac{\eta_\mu'(\theta_*)^2}{\eta_\mu(\theta_*)} > 0. \end{aligned}$$

Strictness holds because $\eta_\mu > 0$ implies $\eta_\mu' \ne 0$. On the two-dimensional slice $b = e^t b_*$, the function

$$\Psi(a, t) = e^{2\operatorname{Im} t}\big(Q(a, e^t b_*) + \widetilde{\eta}(\theta_* + 2\operatorname{Re} t)\big)$$

has positive definite Levi form at $(a_*, 0)$, by the calculation (3.1) with η_μ replaced by $\widetilde{\eta}$. On the nearby part of S it satisfies

$$\Psi = e^{2\operatorname{Im} t}\eta_\mu^\alpha(F - C) \le 0, \qquad \Psi(a_*, 0) = 0.$$

The coordinate-disc lemma [7, Lemma 2.5], applied in that slice and in a neighborhood contained in X_μ, gives a coordinate-line disc through p_* whose boundary satisfies $\Psi > 0$. Coordinates on the slice extend to ambient holomorphic coordinates. Since $\Psi \le 0$ on S near p_*, the boundary of the disc lies in $\mathcal{G}(X_\mu)$. Item (iv) in Section 2 gives $p_* \in \mathcal{G}(X_\mu)$, contradicting $p_* \in S$. Therefore $\mathcal{G}(X_\mu) = X_\mu$, and item (i) proves that X_μ is Oka. □

For part (ii), we use the following continuity-principle consequence.

Lemma 3.2. *Let $n \geq 2$. If $f : \mathbb{C} \to \mathbb{C}^n$ is nonconstant and $Z = \overline{f(\mathbb{C})}$, then every connected component of $\mathbb{C}^n \setminus Z$ satisfies the continuity principle in* [19, Theorem 3.2(5)] *for continuous families of analytic sets whose local dimensions are at least $n-1$.*

Proof. By the strict-peak characterization [21, Proposition 2.3(iv)] (see also [2, Proposition 3.12]), a closed set fails the local maximum property precisely when a smooth strictly plurisubharmonic function has a strict local maximum on it. Suppose this happens for Z. Then there exist a point $q \in Z$, an open neighborhood V of q, and such a function ϕ on V satisfying, after subtraction of $\phi(q)$,

$$\phi(q) = 0, \qquad \phi < 0 \quad \text{on } (Z \cap V) \setminus \{q\}.$$

Choose a ball $B' \Subset V$ containing q. Since $Z \cap \partial B'$ is compact and does not contain q, there are $\epsilon > 0$ and a neighborhood N of $\partial B'$ in V such that $\phi \leq -2\epsilon$ on $Z \cap N$. Hence

$$v(\zeta) = \begin{cases} \max\{\phi(f(\zeta)), -\epsilon\}, & f(\zeta) \in B', \\ -\epsilon, & f(\zeta) \notin B' \end{cases}$$

is a well-defined subharmonic function on $\mathbb{C}$. Indeed, the two expressions agree with the constant $-\epsilon$ near $f^{-1}(\partial B')$, so the assertion follows from the pasting lemma.

The function v is bounded above by zero. Since $q \in Z = \overline{f(\mathbb{C})}$ and ϕ is continuous, one has $\sup_{\mathbb{C}} v = 0$. At least one component of the nonconstant map f is a nonconstant entire function and is therefore unbounded. Hence $f(\mathbb{C})$ is not contained in B', and v takes the value $-\epsilon$. This contradicts the Liouville theorem for bounded-above subharmonic functions. Thus Z has the local maximum property.

By the local maximum theorem [21, Theorem 4.2] (see also [20, Proposition 2.2]), the closed set Z is 1-pseudoconcave. Equivalently, in the convention of [19, Theorem 3.2], $\mathbb{C}^n \setminus Z$ is Hartogs 1-pseudoconvex. Every connected component inherits this property: the image of the filled polydisc in the definition is connected, contains the image of the Hartogs figure, and remains in $\mathbb{C}^n \setminus Z$. The desired continuity principle is [19, Theorem 3.2(5)]. □

Proof of Theorem 3.1(ii). Take $\mu = \pi$. At $p = (-1/2, e_1)$, one has $\theta = 0$ and $\rho_\pi(p) = 5/4 > 0$. Suppose that a nonconstant entire map $f : \mathbb{C} \to \mathbb{C}^n$ with $f(0) = p$ had positive distance from K_π. The closure $Z = \overline{f(\mathbb{C})}$ of its image would be disjoint from K_π. Since K_π is connected, it is contained in a single connected component U of $\mathbb{C}^n \setminus Z$. Compactness then shows that U contains a neighborhood of K_π.

For $s \in [0,1]$, consider the analytic hypersurfaces with boundary

$$A_s = \{(a,b) : a = -s/2,\ e^{-\pi/2} < \|b\| < e^{\pi/2}\}.$$

Their closures depend continuously on s in the Hausdorff topology, and the closure of $\bigcup_{s\in[0,1]} A_s$ is compact. Furthermore, $\overline{A}_0 \subset K_\pi$. On either boundary sphere, $\theta = \pm\pi$ and

$$\rho_\pi(-s/2, b) = |1 - s/2|^2 - 1 \leq 0.$$

Thus all boundaries lie in $K_\pi \subset U$. Let I be the set of $s \in [0,1]$ such that $\overline{A}_t \subset U$ for every $0 \leq t \leq s$. The set I contains 0 and is relatively open by compactness and Hausdorff continuity. If $s = \sup I$, then the boundaries of all A_t, $0 \leq t \leq s$, lie in $K_\pi \subset U$. Lemma 3.2, applied to the family on $[0,s]$, gives $\overline{A}_s \subset U$. Hausdorff continuity then shows that either $s = 1$ or I contains points larger than s. Hence $I = [0,1]$. This contradicts $p \in A_1 \cap Z$. Therefore $e_{K_\pi}(p) = 0$. The Oka property already proved, together with Proposition 2.6, gives $H^-(K_\pi) = K_\pi$. □

The worm construction proves the first converse failure in Theorem 1.1. The same clearance criterion also reaches compact domains that do not satisfy the hypotheses of the polynomial-convexity, tangential Levi-form, or periodic-transport results described in the Introduction, as the following example shows.

3.2. **The Real-Algebraic Compact Set.** For $n \geq 3$, use coordinates $(z, t, w) \in \mathbb{C} \times \mathbb{C}^{n-2} \times \mathbb{C}$ and write

$$R = |z|^2, \qquad T = \|t\|^2.$$

Consider

$$\rho(z, t, w) = 16(R-1)^2 - T + T^2 + |1 + zw|^2 - 1, \qquad K_\rho = \{\rho \leq 0\}.$$

Theorem 3.3. *For every $n \geq 3$, the set K_ρ is the closure of a connected bounded domain with smooth real-algebraic boundary. Its boundary has a strictly positive tangential Levi eigenvalue at every point. At every point of*

$$\Gamma = \{(z, 0, 0) : |z| = 1\}$$

the tangential Levi form has one positive and $n-2$ negative eigenvalues. Moreover,

$$H(K_\rho) = K_\rho, \qquad \mathbb{C}^n \setminus K_\rho \text{ is Oka},$$

although K_ρ is not polynomially convex.

Proof. The set K_ρ is contained in

$$16(R-1)^2 + (T - \tfrac{1}{2})^2 + |1 + zw|^2 \leq \tfrac{5}{4}.$$

With $R_\pm = 1 \pm \sqrt{5}/8$, this gives

$$\tfrac{1}{2} < R_- \leq |z|^2 \leq R_+ < 2, \qquad T \leq \tfrac{1+\sqrt{5}}{2} < 2, \tag{3.7}$$

together with a uniform bound on w. Hence K_ρ is compact and is separated from $z = 0$.

On $\{z \neq 0\}$, use the variables

$$\zeta = z/|z|, \qquad x = |z|^2 - 1, \qquad v = 1 + zw.$$

The map $(z, t, w) \mapsto (\zeta, x, t, v)$ is a diffeomorphism from $\mathbb{C}^* \times \mathbb{C}^{n-2} \times \mathbb{C}$ onto $S^1 \times (-1, \infty) \times \mathbb{C}^{n-2} \times \mathbb{C}$; its inverse is $z = \sqrt{1+x}\,\zeta$ and $w = (v-1)/z$. In these variables, the defining function is

$$16x^2 - \|t\|^2 + \|t\|^4 + |v|^2 - 1.$$

Its critical equations are $x = v = 0$ and $(2T - 1)t = 0$; the corresponding critical values are -1 and $-5/4$. Thus the zero level is regular. It follows that $K_\rho = \overline{\operatorname{int} K_\rho}$ and that ∂K_ρ is a smooth real-algebraic hypersurface. For fixed ζ, join (x, v, t) to $(0, 0, 0)$ by $(x, v, t) \mapsto (rx, rv, rt)$, $0 \leq r \leq 1$. With $s = r^2$, the defining expression along this path is

$$s(16x^2 + |v|^2 - T) + s^2 T^2 - 1.$$

This convex quadratic is negative at $s = 0$ and nonpositive at $s = 1$, so it is nonpositive for $0 \leq s \leq 1$. Thus every fixed-ζ slice of K_ρ is star-shaped with respect to $(x, v, t) = (0, 0, 0)$. In the original coordinates, these centers form the circle

$$\{(z, 0, -1/z) : |z| = 1\} \subset \operatorname{int} K_\rho.$$

The same argument with strict inequalities applies to the interior. Therefore both K_ρ and its interior are connected.

On the complex (z, w)-plane,

$$\mathcal{L}_\rho(\xi_z, 0, \xi_w) = 32(2R-1)|\xi_z|^2 + |w\xi_z + z\xi_w|^2. \tag{3.8}$$

Its determinant is $32R(2R-1) > 0$ on the region (3.7). Hence the Levi form is positive definite on the complex two-dimensional subspace $\{\xi_t = 0\}$. This subspace intersects every boundary complex tangent hyperplane in a nonzero complex direction, on which the Levi form is positive. On Γ,

$$\partial\rho = z\,dw, \qquad \mathcal{L}_\rho(\xi_z, \xi_t, 0) = 32|\xi_z|^2 - \|\xi_t\|^2.$$

The complex tangency condition is $\xi_w = 0$, so the last display proves that the tangential Levi form has one positive and $n-2$ negative eigenvalues on Γ.

The circle $\Gamma \subset K_\rho$ gives

$$\{(z,0,0) : |z| \le 1\} \subset \widehat{K}_\rho, \qquad (0,0,0) \notin K_\rho,$$

by the one-variable maximum principle. Thus K_ρ is not polynomially convex.

It remains to prove the clearance-hull equality. If $|z|^2 < R_-$ or $|z|^2 > R_+$, the whole (t,w)-fiber is disjoint from K_ρ. Every point in such a fiber lies on an affine complex line in the w direction that is disjoint from the compact set K_ρ. Since the line is closed, its distance from K_ρ is positive. Hence

$$H(K_\rho) \subset \{R_- \le |z|^2 \le R_+\}.$$

If $S = H(K_\rho) \setminus K_\rho$ were nonempty, then ρ would attain a positive maximum on the compact set $H(K_\rho)$ at a point $p \in S$. Equation (3.8) gives two positive Levi eigenvalues at p. The coordinate-disc lemma [7, Lemma 2.5], applied in a neighborhood contained in $\mathbb{C}^n \setminus K_\rho$, gives a coordinate-line disc D centered at p such that $\rho|_{\partial D} > \rho(p)$. By maximality, $\partial D \cap H(K_\rho) = \varnothing$. Lemma 2.4 then gives $p \notin H(K_\rho)$, a contradiction. Thus $H(K_\rho) = K_\rho$, and Theorem 2.2 proves that the complement is Oka. □

The set K_ρ is not polynomially convex, so it does not satisfy the hypothesis of [17, Corollary 1.3] or that of the Fatou–Bieberbach-family theorem [12, Theorem 1.1]. On Γ, its tangential Levi form has only one nonnegative eigenvalue, so the hypothesis of [7, Theorem 1.1] is not satisfied either. The nonempty compact set K_ρ cannot be a periodic obstacle $q^{-1}(C)$ from [6, Theorem 1.1], where C is a nonempty compact $\mathscr{O}((\mathbb{C}^*)^n)$-convex subset of $(\mathbb{C}^*)^n$ and $q(z) = (e^{iz_1}, \dots, e^{iz_n})$, since every such obstacle is invariant under $2\pi\mathbb{Z}^n$ and hence unbounded. Nor is it a convex tube $\mathbb{R}^n + iB$, with $B \subset \mathbb{R}^n$ compact and convex, as in [6, Theorem 1.2(i)]. Finally, it is not a complex-affine image of a Cartesian product $K_1 \times \cdots \times K_n$ of planar compact sets, the class covered by [6, Theorem 1.2(ii)]. Indeed, a compact product with nonempty interior has at least two planar factors with nonempty interiors. If its boundary were C^1, fixing interior points in all but one factor would show that each planar boundary is C^1. At a point with boundary coordinates in two factors, the corresponding boundary faces would then meet with linearly independent normal directions, a contradiction. The same obstruction persists under complex-affine changes of coordinates. Consequently, Theorem 3.3 is an example for which the clearance equality proves the Oka property while none of these cited sufficient results applies.

4. A Non-Oka Complement with Polynomially Convex Inner Parallel Sets

The second implication in (1.3) cannot be reversed for arbitrary compact regular closed sets. The construction below gives a connected compact set with connected complement. Every nonempty inner parallel set is polynomially convex and hence has Oka complement, while the complement of the original compact set is not Oka. Smoothness of the boundary is not asserted.

Lemma 4.1. *Let $n \ge 1$, and suppose $A, B \subset \mathbb{C}^n$ are compact and polynomially convex. If a polynomial P satisfies*

$$|P| \le \eta \quad \text{on } A, \qquad |P-1| \le \eta \quad \text{on } B, \qquad \eta = \frac{1}{16},$$

then $A \cup B$ is polynomially convex.

Proof. The polynomial hulls of $P(A)$ and $P(B)$ are contained, respectively, in the disjoint discs

$$\overline{\Delta(0,\eta)} \qquad \text{and} \qquad \overline{\Delta(1,\eta)}.$$

The polynomial separation form of Kallin's lemma [3] therefore gives

$$\widehat{A \cup B} = \widehat{A} \cup \widehat{B} = A \cup B.$$

□

Theorem 4.2. *For every $n \geq 2$, there is a connected compact regular closed set $K \subset \mathbb{C}^n$ whose complement $X = \mathbb{C}^n \setminus K$ is connected and has the following properties:*

(i) *For every $\delta > 0$ for which*

$$K_{-\delta} = \{z \in K : \operatorname{dist}(z, X) \geq \delta\}$$

is nonempty, it is a finite union of pairwise disjoint closed Euclidean balls, possibly including one-point balls, and this union is polynomially convex. Moreover, $H^-(K) = K$.

(ii) *There is a nonempty open set $W \subset X$ such that, for every $p \in W$ and every holomorphic map $f : \mathbb{C} \to X$ with $f(0) = p$, the map f is constant. Consequently X is neither Oka nor Oka-1.*

The interior of K has infinitely many connected components, and ∂K is not a smooth hypersurface.

Proof. Step 1: Construction and regular closedness. Let

$$\Sigma = \{z \in \mathbb{C}^n : \|z\| = 1\}, \qquad q = (1, 0, \ldots, 0), \qquad \Sigma_+ = \{z \in \Sigma : \operatorname{Re} z_1 \geq \tfrac{1}{2}\}.$$

The compact spherical cap Σ_+ is connected and has no isolated points. Choose a sequence of distinct points $s_j \in \Sigma_+$ which is dense in Σ_+. We shall select radii

$$0 < r_j < 2^{-j-4}$$

so that the closed balls

$$D_j = \overline{B((1 + r_j)s_j, r_j)}$$

are pairwise disjoint and every finite union $A_j = D_1 \cup \cdots \cup D_j$ is polynomially convex. Choose any $0 < r_1 < 2^{-5}$; the ball $A_1 = D_1$ is polynomially convex. Suppose that $r_1, \ldots, r_{j-1}$ have been selected, where $j \geq 2$. Each D_i meets Σ only at s_i, so $s_j \notin A_{j-1}$. Polynomial convexity gives a polynomial Q such that $|Q(s_j)| > \sup_{A_{j-1}} |Q|$. Set

$$P_j = \frac{Q}{Q(s_j)}, \qquad \alpha = \sup_{A_{j-1}} |P_j| < 1,$$

choose $N \in \mathbb{N}$ with $\alpha^N < 1/16$, and let $P = P_j^N$. Then

$$P(s_j) = 1, \qquad \sup_{A_{j-1}} |P| < \tfrac{1}{16}.$$

As $r \downarrow 0$, the closed ball $\overline{B((1 + r)s_j, r)}$ shrinks to $\{s_j\}$. Choosing r_j sufficiently small, while retaining $r_j < 2^{-j-4}$, makes $|P - 1| < 1/16$ on D_j and makes D_j disjoint from A_{j-1}. Lemma 4.1 proves that A_j is polynomially convex and completes the induction. Notice that D_j is externally tangent to Σ at s_j.

Set

$$K = \Sigma_+ \cup \bigcup_{j \geq 1} D_j, \qquad \Omega = \bigcup_{j \geq 1} \operatorname{int} D_j.$$

Each D_j lies in $\{\|z\| \geq 1\}$, meets Σ_+ at s_j, and has diameter $2r_j \to 0$. Every accumulation point of balls with indices tending to infinity is in Σ_+. Thus K is compact. It is connected because every ball meets the connected cap Σ_+.

The family $\{D_j\}$ is locally finite outside Σ_+. A point of Σ_+ cannot be interior to K because every neighborhood meets the open unit ball, which is disjoint from K. Away from Σ_+, a boundary point of any D_j is not interior to the pairwise disjoint, locally finite union. Hence $\operatorname{int} K = \Omega$. Since $D_j = \overline{\operatorname{int} D_j}$ and the points s_j are dense in Σ_+,

$$K = \overline{\Omega} = \overline{\operatorname{int} K}. \tag{4.1}$$

In particular, K is regular closed but its interior is the disconnected union of the open balls $\operatorname{int} D_j$.

The boundary is not a topological hypersurface. Indeed, choose an index j such that s_j belongs to the relative interior $\{z \in \Sigma : \operatorname{Re} z_1 > 1/2\}$ of Σ_+. A neighborhood of s_j in Σ and a neighborhood of s_j in ∂D_j are two embedded real $(2n-1)$-dimensional submanifolds contained in ∂K, and their intersection is $\{s_j\}$. If ∂K were a topological $(2n-1)$-manifold near s_j, invariance of domain would make both submanifolds open in ∂K near s_j. Their intersection would then make $\{s_j\}$ open in ∂K, which is impossible. Hence ∂K is not a smooth hypersurface.

Step 2: The inner parallel sets and the inner clearance hull. For $x \in \operatorname{int} D_j$, the distance to X is exactly the distance to ∂D_j:

$$\operatorname{dist}(x, X) = r_j - \|x - (1 + r_j)s_j\|.$$

The lower bound follows from $D_j \subset K$. For the reverse bound, every point of $\partial D_j \setminus \{s_j\}$ is approached by points of X: the other balls are locally finite there, and no other ball meets D_j. The tangency point s_j is approached from the open unit ball. Every point of Σ_+ has distance zero from X. Therefore

$$K_{-\delta} = \bigcup_{r_j \geq \delta} \overline{B((1 + r_j)s_j, r_j - \delta)}, \qquad \delta > 0.$$

Only finitely many indices occur. The sets on the right are polynomially convex. Indeed, insert them in increasing order of their indices. At the jth insertion the new ball is contained in D_j, whereas the union of the preceding balls is contained in A_{j-1}. The polynomial used to construct D_j therefore still satisfies the two estimates in Lemma 4.1. Induction proves the claim. Thus the complement of every nonempty $K_{-\delta}$ is Oka by Kusakabe's Corollary 1.3 [17].

More generally, let $L \Subset \Omega$ be compact. The open sets $\operatorname{int} D_j$ form a pairwise disjoint cover of Ω, so compactness of L gives finitely many indices j for which $L \cap D_j \neq \varnothing$. Within each such ball, choose a concentric closed ball $D'_j \Subset \operatorname{int} D_j$ containing $L \cap D_j$. The preceding induction, with D'_j in place of the inner parallel ball, shows that $\bigcup_j D'_j$ is polynomially convex. The inclusion $H(L) \subset \widehat{L}$, proved in Corollary 2.3, gives

$$H(L) \subset \widehat{L} \subset \bigcup_j D'_j \Subset \Omega.$$

Together with (4.1) and $L \subset H(L)$, this shows

$$H^-(K) = \overline{\bigcup_{\varnothing \neq L \Subset \Omega} H(L)} = K.$$

Step 3: Connectedness of the complement. The exterior region

$$E = \{\|z\| > 1\} \setminus \bigcup_{j \geq 1} D_j$$

is path connected. Join two given points by a polygonal path γ in $\{\|z\| > 1\}$. Its compact image has distance $d > 0$ from Σ. The $d/3$-neighborhood of this image meets only finitely many balls: if it meets D_j, then D_j contains a point at distance at least $2d/3$ from Σ, and hence $2r_j \geq 2d/3$. After a perturbation within this neighborhood that fixes the endpoints, the path is transverse to the boundaries of all these balls and has only finitely many crossings.

For each component (a, b) of $\gamma^{-1}(\operatorname{int} D_j)$, the points $\gamma(a)$ and $\gamma(b)$ lie in $\partial D_j \setminus \{s_j\}$. This punctured sphere is path connected because its real dimension is $2n - 1 \geq 3$, so the two points can be joined by an arc β in it. The image of the arc is compact and disjoint

from Σ. Local finiteness away from Σ, together with pairwise disjointness of the balls, gives it positive distance from every other ball. If $c_j = (1 + r_j)s_j$ is the center of D_j, then

$$\beta_\epsilon(t) = c_j + (1 + \epsilon)(\beta(t) - c_j)$$

lies outside D_j for every $\epsilon > 0$. For sufficiently small ϵ, choose $a' < a < b < b'$ so close to a, b that $\gamma(a')$ and $\gamma(b')$ can be joined to the respective endpoints of β_ϵ inside the exterior tubular neighborhood of $\partial D_j \setminus \{s_j\}$. These two connectors and β_ϵ avoid all the balls and remain in $\{\|z\| > 1\}$. Replacing $\gamma|_{[a',b']}$ by this path for each of the finitely many crossing components gives a path in E. Thus E is path connected.

Every point of $\Sigma \setminus \Sigma_+$ has a small neighborhood disjoint from K, connecting it to the open unit ball. Since $r_j < 1/32$ and $\mathrm{Re}((s_j)_1) \geq 1/2$, every $z \in D_j$ satisfies

$$\mathrm{Re}\, z_1 \geq (1 + r_j)\,\mathrm{Re}((s_j)_1) - r_j \geq \frac{1 - r_j}{2} > \frac{15}{32}.$$

Thus the negative real z_1-axis is disjoint from K, and it joins the open unit ball to E through the point $(-1, 0, \ldots, 0) \in \Sigma \setminus \Sigma_+$. Thus X is path connected.

Step 4: Entire-curve rigidity in the complement. Consider

$$h(z) = \mathrm{Re}\, z_1 - 1 + \tfrac{1}{4}\|z - q\|^2, \qquad \kappa = \tfrac{1}{8}.$$

Inside the open unit ball,

$$h(z) = \tfrac{1}{2}(\|z\|^2 - 1) - \tfrac{1}{4}\|z - q\|^2 < 0, \qquad \mathcal{L}_h(v) = \tfrac{1}{4}\|v\|^2 \quad (v \in \mathbb{C}^n).$$

Also $\Sigma \cap B(q, 1) \subset \Sigma_+$ because $\|z - q\|^2 = 2 - 2\,\mathrm{Re}\, z_1$ on Σ. Consequently the function

$$u(z) = \begin{cases} \max\{h(z), -\kappa\}, & \|z\| < 1,\ \|z - q\| < 1, \\ -\kappa, & \text{otherwise}, \end{cases} \qquad z \in X,$$

is well defined, continuous, and plurisubharmonic. The only gluing boundary in X for the first region is $\|z - q\| = 1$; there $h < -1/4 < -\kappa$, so the definitions agree in a neighborhood of every such point. The other boundary, $\Sigma \cap B(q, 1)$, is contained in K. Hence

$$-\kappa \leq u < 0 \quad \text{on } X.$$

The set

$$W = \{z \in X : u(z) > -\kappa\}$$

is nonempty: $u((1 - t)q) = -t + t^2/4 > -\kappa$ for all sufficiently small $t > 0$. Moreover $W \subset B_1 = B(0, 1)$. If a holomorphic map $f : \mathbb{C} \to X$ satisfies $f(\zeta_0) \in W$ for some $\zeta_0 \in \mathbb{C}$, the bounded subharmonic function $u \circ f$ is constant, with value greater than $-\kappa$. Therefore $f(\mathbb{C}) \subset W \subset B_1$. Every coordinate of f is bounded, so Liouville's theorem makes f constant. Choose $p \in W$ and $x \in X \setminus \{p\}$. Let $\ell : [0, 1] \to X$ be a path from p to x, and choose a continuous function $\lambda : \mathbb{C} \to [0, 1]$ that equals 0 on a neighborhood of 0 and 1 on a neighborhood of 1. The map $g = \ell \circ \lambda : \mathbb{C} \to X$ is holomorphic near 0 and 1, since it is locally constant there, and satisfies $g(0) = p$ and $g(1) = x$. If X were Oka-1, Definition 1.1 in [1], applied with the open Riemann surface $\mathbb{C}$, the empty Runge approximation set, the discrete interpolation sequence $(0, 1)$, and interpolation orders $k_1 = k_2 = 1$, would give an entire map $F : \mathbb{C} \to X$ with $F(0) = p$ and $F(1) = x$. This contradicts the preceding assertion for entire maps meeting W, so X is not Oka-1. Since X is connected, the entire-curve rigidity at p also shows directly that X is not Oka by [7, Lemma 2.1].

This proves all assertions of Theorem 4.2. □

Theorem 4.2 disproves the second converse in (1.3): every nonempty $K_{-\delta}$ is polynomially convex and hence has Oka complement by [17, Corollary 1.3], whereas $\mathbb{C}^n \setminus K$ is not Oka. Since K has neither smooth boundary nor connected interior, the following question remains open.

Question 4.3. If $K \Subset \mathbb{C}^n$, $n \geq 2$, is the closure of a bounded domain with C^2 boundary and $H^-(K) = K$, must $\mathbb{C}^n \setminus K$ be Oka?

Acknowledgments. The author is deeply grateful to his advisor, Professor Song-Yan Xie, for his help and support. He also thanks Professor Yuta Kusakabe for asking him questions concerning Oka complements of compact sets.

GPT-6-Astra was used to assist in the preparation and revision of this manuscript. The author independently checked every mathematical statement and assumes full responsibility for the content of the paper.

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